

Introduction & Data Model

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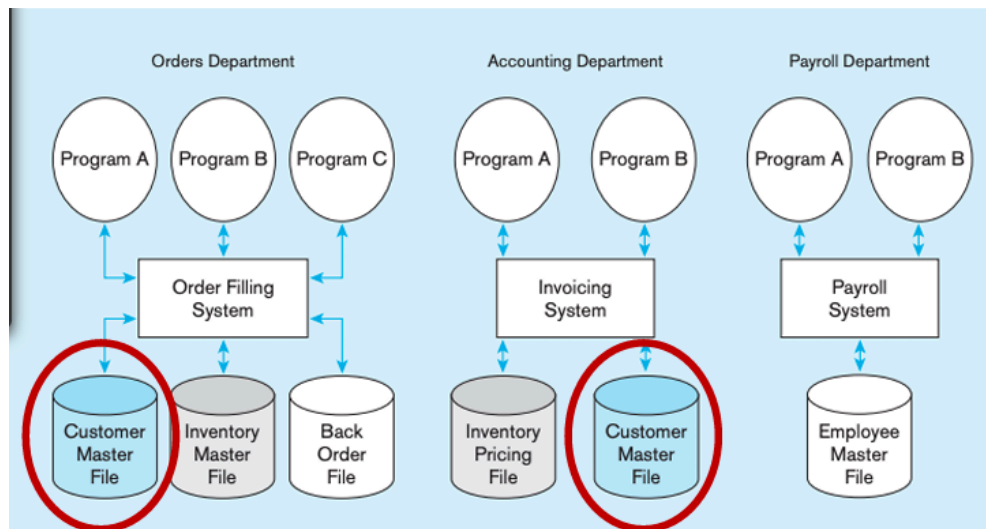
Introduction

Definition

- **Database** is an organized collection of logically related data (terorganisasi dengan baik, dan saling terhubung)
- **Data** is a stored representations of meaningful objects and events.
 - Structured, i.e. numbers, text, dates
 - Unstructured, i.e. images, video, documents
- **Information** is data processed to increase knowledge in the person using the data
 - **Data in context** helps users understand data
- **Metadata** is data that describes the properties or characteristics of user data, including data types, field sizes, allowable values, and data context

File Processing System vs The Database Approach

Traditional File Processing System



When computer-based data processing was first available, there were no databases → computers had to store, manipulate, and retrieve large files of data.

Disadvantages of File Processing

- **Program-data dependence**
All programs maintain metadata for each file they use
- **Problems:**
 - each application programmer must maintain their own data

- each application program needs to include code for the metadata of each file
- each application program must have its own processing routines for reading, inserting, updating, and deleting data
- lacks coordination and central control
- non-standard file formats

- **Data redundancy (duplication of data)**

Different systems/programs have separate copies of the same data

Problems:

- waste of space
- causes more maintenance headaches
- **biggest problem:** when data changes in one file, inconsistencies may happen, which compromises data integrity

- **Limited data sharing**

No centralized control of data

- **Lengthy development times**

Programmers must design their own file formats

- **Excessive program maintenance**

80% of information systems budget

- **Difficulty in accessing data**

Need to write a new program to carry out each new task

- **Data isolation**

Multiple files and formats

- **Integrity problems**

Integrity constraints become “buried” in program code rather than being stated explicitly. Hard to add new constraints or modify existing ones.

- **Atomicity of updates**

Failures may leave database in an inconsistent state with partial updates carried out

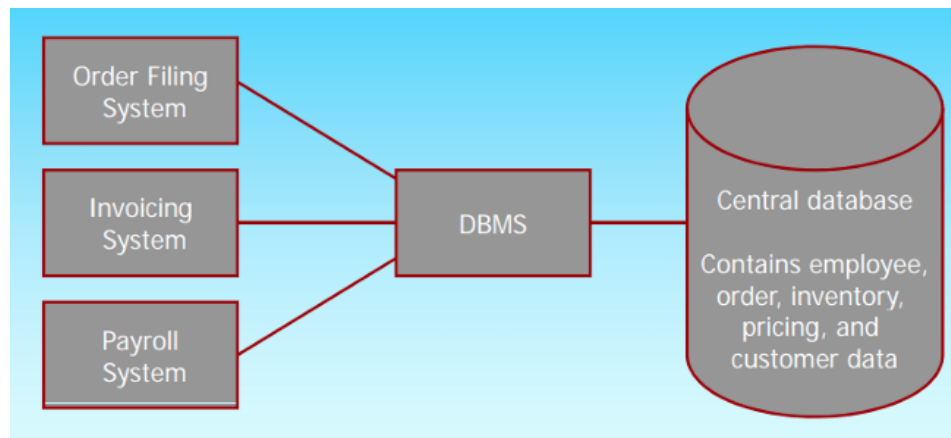
- **Concurrent access by multiple users**

Concurrent access needed for performance. Uncontrolled concurrent accesses can lead to inconsistencies

- **Security problems**

Hard to provide user access to some, but not all, data

Database Management System (DBMS)



DBMS is a software system that is used to create, maintain, and provide controlled access to user databases. DBMS manages data resources like an operating system manages hardware resources.

Advantages/Roles of Database Approach

- Improved data sharing → different users get different views of the data
- Enforcement of standards → all data access is done in the same way
- Improved data quality → constraints, data validation rules
- Better data accessibility/responsiveness → use of standard data query language (SQL) → SQL bersifat deklaratif
- Security, backup/recovery, concurrency → disaster recovery is easier

Costs and Risks of the Database Approach

- Up-front costs
 - Installation management cost and complexity
 - Conversion costs
- Ongoing costs
 - requires new, specialized personnel
 - needs explicit backup and recovery
- Organizational conflict
 - old habits die hard (?)

Database Application

Components of The Database Environment

- CASE Tools: computer-aided software engineering

- Repository: centralized storehouse of metadata
- Database Management System (DBMS): software for managing the database
- Database: storehouse of the data
- Application programs: software using the data
- User Interface: text and graphical displays to users
- Data administrators: personnel responsible for maintaining the database
- System developers: personnel responsible for designing databases and software
- End users: people who use the applications and databases

The Range of Database Application

- Personal databases
- Workgroup databases
- Departmental/divisional databases
- Enterprise database
 - Enterprise resource planning (ERP) systems
 - Data warehousing implementations

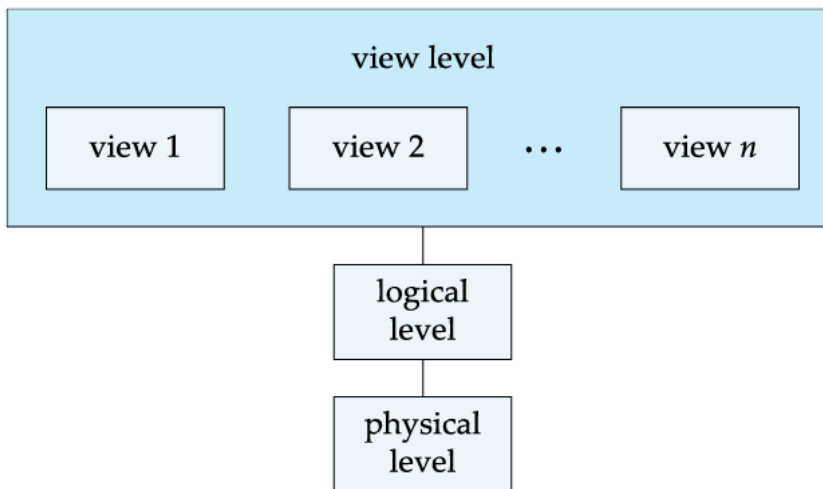
Evolution of Database Technologies

Levels of Abstraction

1. **Physical level:** describes how a record is stored, i.e. waktu tubes AP1 & AP2 itu kita langsung berurusan dengan tipe datanya → we work on a physical level
2. **Logical level:** describes data stored in database, and the relationships among the data
3. **View level:** application programs hide details of data types. Views can also hide information for security purposes. seolah-olah ada basis data lain untuk user tipe lain karena viewnya berbeda, padahal itu manipulasi access saja.

View of Data

An architecture for a database system (ANSI/SPARC Architecture)



Note: kalau disuruh gambar ini diagramnya, kalau disuruh jelasin level abstraksinya yg sebelum ini (level of abstraction)

Physical Data Independence

The ability to modify the physical schema without changing the logical schema.

- Applications depend on logical schema
- In general

Instances and Schema

1. Schema: structure of the database
 - a. Logical schema
 - b. Physical schema
2. Instance: the actual content of the database at a particular point in time

Data Definition Language (DDL)

Jika mempengaruhi skema, ia akan masuk DDL. DDL compiler generates a set of table templates stored in a data dictionary, which contains metadata (data about data)

- Database schema
- Integrity constraints
 - Primary key (ID uniquely identifies instructors)
- Authorization

Data Manipulation Language (DML)

Language for accessing and updating the data organized by the appropriate data model.

- Procedural DML: require a user to specify what data are needed and how to get those data
- Declarative DML: paling umum digunakan. require a user to specify what data are needed without specifying how to get those data.

SQL query is nonprocedural. A query takes as input several tables (possibly only one) and always returns a single table.

Database Design

- Logical design: deciding on the database schema. Database design requires that we find a "good" collection of relation schemas.
 - Business decision
 - Computer science decision
- Physical design: deciding on the physical layout of the database

Database Management Systems

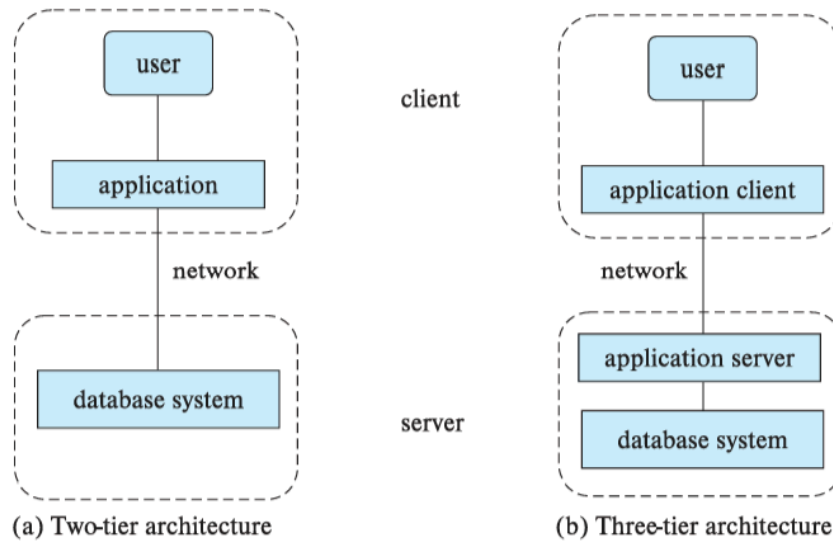
A DBMS is partitioned into modules that deal with each of the responsibilities of the overall system. The functional components of a DBMS can be divided into:

- The storage manager
- The query processor component
- The transaction management component: termasuk ke dalam storage manager, ia mengurus data tetapi tidak mengurus physicalnya

The Database System Environment

Database system refers to an organization of components that define and regulate the collection, storage, management, and use of data within a database environment. The database system is composed of the five major parts: hardware, software, people, procedures, and data.

Database Applications Architecture



- **Two-tier architecture** — the application resides at the client machine, where it invokes database system functionality at the server machine
- **Three-tier architecture** — the client machine acts as a front end and does not contain any direct database calls.

Benefits of client-server/three-tier architecture:

Kita tinggal membuat satu backend, sebagai logika dasarnya, tetapi bisa digunakan/diakses dengan berbagai device (mobile, desktop, etc.).

Data Model

Data Modelling

Data modelling is a technique aimed at optimizing the way that information is stored and used within an organization (menghindari redundansi, data yang memungkinkan bias, dsb.)

The process of data modelling:

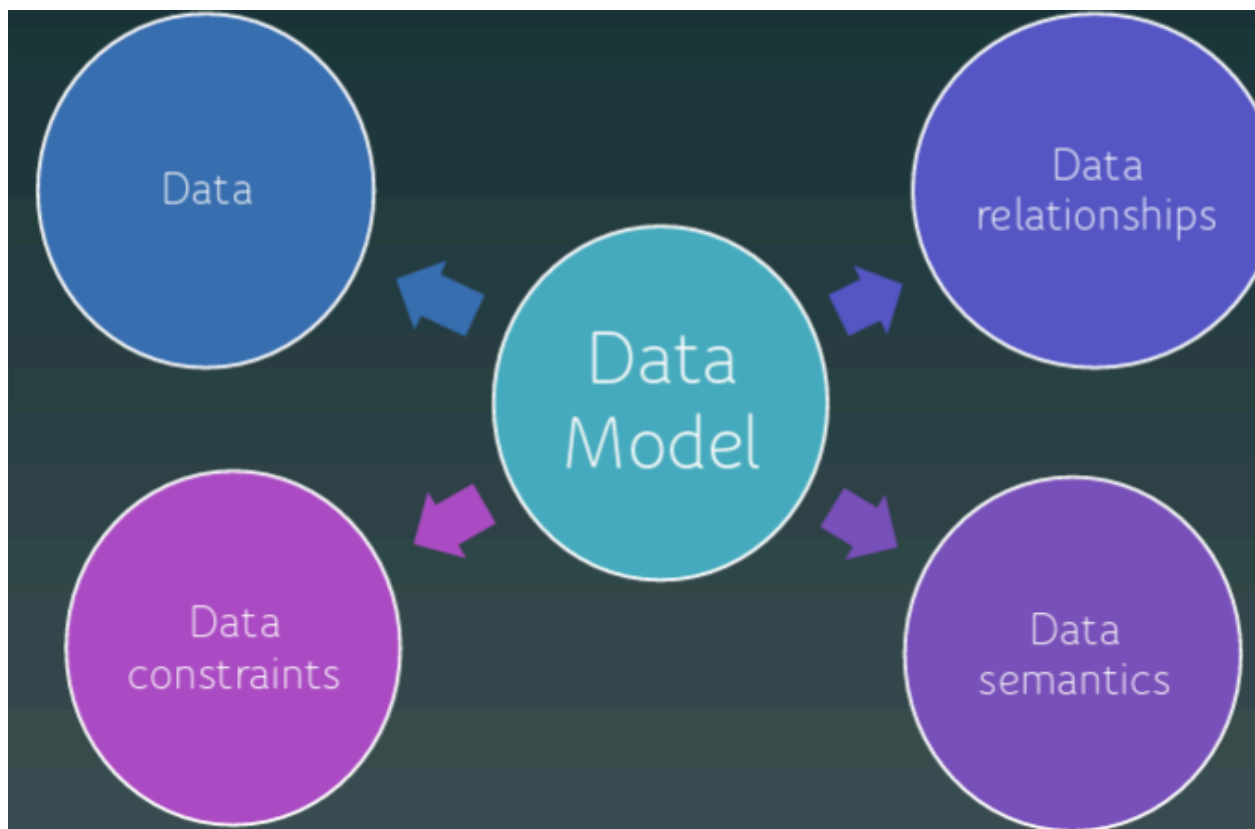
1. identification of the main data groups
i.e. ada data grup dosen dan grup mahasiswa
2. defining the detailed content of each of these groups
i.e. di data dosen ada komponen apa aja
3. result: structured definitions for all of the information that is stored and used within a given system

An essential precursor to analysis, design, maintenance & documentation, and improving the performance of an existing system.

There are three different types of data models produced while progressing from requirements to the actual database.

- **Conceptual data model:** a set of technology independent specifications about the data and is used to discuss initial requirements with the business stakeholders.
 - Tujuan conceptual data model adalah untuk komunikasi dengan user, memastikan bahwa pemahaman terhadap user requirement
- **Logical data model:** the structures of the data that can be implemented in databases. Di logical data model, sudah mempertimbangkan teknologi yang digunakan (such as object oriented, document oriented, relational, etc.)
- **Physical data model:** organizes the data into tables, and accounts for access, performance and storage details.

Data model



Data: bagaimana kita menstrukturkan data

Data relationships: bagaimana hubungan antar data

Data constraints: constraint apa saja yang di-enforce thd data

Data semantics: jika tidak ada, org" tidak akan memahami.

Types of data model:

- Level conceptual:
 - Entity-relationship model
 - Relational model
- Level logical:
 - Object-based data model (object oriented and object relational)
 - Semi-structured data model (XML)
- Level physical:
 - Network model
 - Hierarchical model

Entity Relationship Model

Relational Model

- Consist of collection of tables to represent data and the relationship among those data
- **Example of tabular data in the relational model**

<i>customer_id</i>	<i>customer_name</i>	<i>customer_street</i>	<i>customer_city</i>	<i>account_number</i>
192-83-7465	Johnson	12 Alma St.	Palo Alto	A-101
192-83-7465	Johnson	12 Alma St.	Palo Alto	A-201
677-89-9011	Hayes	3 Main St.	Harrison	A-102
182-73-6091	Turner	123 Putnam St.	Stamford	A-305
321-12-3123	Jones	100 Main St.	Harrison	A-217
336-66-9999	Lindsay	175 Park Ave.	Pittsfield	A-222
019-28-3746	Smith	72 North St.	Rye	A-201

Schema: judul setiap kolom (atribut)

Hubungan antar tabel direpresentasikan melalui nilai/value

Object-Oriented Data Model

- Adaptations of the object-oriented programming paradigm to database systems
- The object-oriented paradigm is based on encapsulating code and data related to an object into single unit

Object-Relational Data Model

- Struktur dasar masih tabel tetapi bisa punya collection sebagai atribut sebuah tabel. (struktur/dasar adalah relational dengan konsep object, i.e. atributnya adalah object instead of tipe dasar)

Semi Structured Data Model (XML)

Setiap data diberikan tag/dijelaskan. Umumnya untuk pertukaran data. Dulu XML, sekarang umumnya menggunakan JSON

Characteristics:

- organized in semantic entities
- similar entities are grouped together
- entities in same group may not have same attributes
- order of attributes not necessarily important
- not all attributes may be required
- size of same attributes in a group may differ
- type of same attributes in a group may differ

Hierarchical Data Model

Network Model

- Data are represented by collections of records.
 - similar to an entity in the E-R model
 - Records and their fields are represented as record type
- Relationships among data are represented by links
 - Sebuah record bisa memiliki lebih dari satu link, sehingga tidak ada duplikasi